11 Clinical Utility Metrics - Calibration Curves, Net Benefit, Operating Points, Event-Level Detection

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0.1 Introduction

Notebook 07 reports discrimination (AUROC, AUPRC), the PhysioNet Utility Score, calibration intercept and slope, and a ratio-based alert burden. That suite leaves six clinically relevant questions unanswered, and this notebook answers them:

1. **Calibration curves.** Intercept and slope are two numbers summarising a whole curve. The flexible calibration curve is plotted directly, and summarised with the Integrated Calibration Index (ICI), E50, E90 and the scaled Brier score (Austin & Steyerberg, 2019; Van Calster et al., 2019).
2. **Decision-curve / net-benefit analysis.** Notebook 07 evaluated net benefit on a threshold grid of 0.01–0.20. Event prevalence in the person-hour table is roughly 0.2 per cent, so every threshold on that grid is an order of magnitude above the prevalence and the whole curve is trivially below "treat none". Net benefit is recomputed here on a prevalence-appropriate grid (Vickers & Elkin, 2006; Vickers et al., 2019).
3. **PPV at a fixed sensitivity.** Alerting systems are commissioned at an operating point, not at a default 0.5 cut-off. Thresholds are chosen to hit sensitivity 0.50 / 0.80 / 0.90 and PPV, specificity and number needed to evaluate are reported there.
4. **False alarms per patient-day.** The ratio-based burden (false alerts per true alert) is not time-normalised, so it cannot be compared against ward-level alarm-fatigue thresholds. Both alert-hours and distinct alarm *episodes* per patient-day are computed here.

5. **Per-patient (event-level) detection.** All of notebook 07's discrimination is per person-hour. Episode-level sensitivity ("what fraction of septic patients were alerted before onset?"), per-patient PPV and the fraction of non-septic stays that received at least one alert are reported alongside the per-hour figures.
6. **Lead time.** Reported as a full distribution (median, IQR, proportion detected at least 3 h ahead), not a single median buried in a wide table.

Everything is computed on the temporal test set (2017–2019) using the predictions already written by notebooks 05, 06 and 08. No model is refitted.

0.2 Metric Functions

The helpers the rest of the notebook runs on. `counts_at_thresholds()` carries the cost that matters: it sorts the predictions once, then reads every threshold off cumulative sums instead of rescanning the vector per threshold.

0.3 Data Assembly

The same merge as notebook 07: primary-model predictions joined to the comparator predictions on `(stay_id, hour)`. Only the unrestricted evaluation window is used: the treatment-anchored window contains zero event hours (notebook 07, H3), so no operating point can be defined in it.

0.4 Calibration Curves

Predicted risk against observed event rate. The diagonal is perfect calibration. The spline curve is the flexible calibration curve; the points are twenty equal-sized bins of predicted risk with binomial confidence intervals. Axes are restricted to the observed prediction range, because at a 0.2 per cent event rate no model ever predicts a risk near 1.

Table 1: Calibration summary. ICI = Integrated Calibration Index (mean absolute difference between predicted risk and the flexible calibration curve). E50/E90 = median and 90th percentile of that difference. Scaled Brier = 1 - Brier / Brier(null); values at or below 0 mean the model is no better than predicting the prevalence for everyone.

variant	model	Intercept	Slope	ICI	E50	E90	Brier	Brier scaled
A	GBT	-6.049	1.143	0.379000	0.350000	0.577000	0.166500	-89.1792
A	NEWS2	-5.246	0.157	0.220000	0.198000	0.447000	0.073260	-38.6909
A	Primary (pooled logistic)	-0.077	1.013	0.000153	0.000102	0.000393	0.001839	0.0037
B	GBT	-6.059	1.306	0.399000	0.376000	0.572000	0.178800	-93.5482
B	NEWS2	-5.222	0.148	0.220000	0.198000	0.447000	0.073340	-37.7788
B	Primary (pooled logistic)	-0.106	1.055	0.000240	0.000205	0.000488	0.001884	0.0038
C	GBT	-5.295	1.253	0.397000	0.371000	0.593000	0.182100	-42.3814
C	NEWS2	-4.377	0.144	0.212000	0.196000	0.393000	0.071150	-15.9496
C	Primary (pooled logistic)	-0.035	1.084	0.000350	0.000333	0.000420	0.004163	0.0082

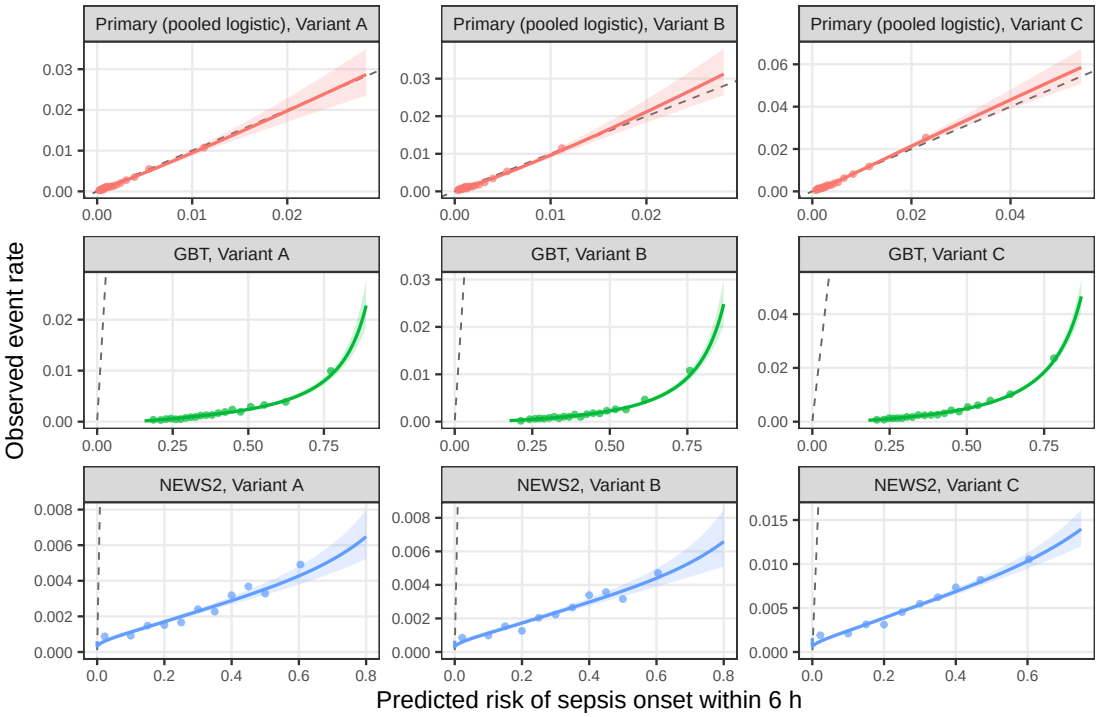


Figure 1: Flexible calibration curves on the temporal test set. The dashed line is perfect calibration, points are twenty equal-sized bins of predicted risk, and the shaded band is the 95 per cent confidence interval. Each panel carries its own scale because the predicted-risk ranges differ by orders of magnitude.

0.5 Decision-Curve Analysis

Net benefit is evaluated on a grid running from one tenth of the event prevalence to roughly ten times it. This is the range in which a sepsis-alerting threshold could plausibly be set: a threshold probability of t encodes a willingness to accept $(1-t)/t$ false alarms per true detection. At a 0.2 per cent event rate, a threshold of 0.01 already encodes a willingness to tolerate only 99 false alarms per detection, which is why the 0.01–0.20 grid used in notebook 07 produced nothing but negative curves.

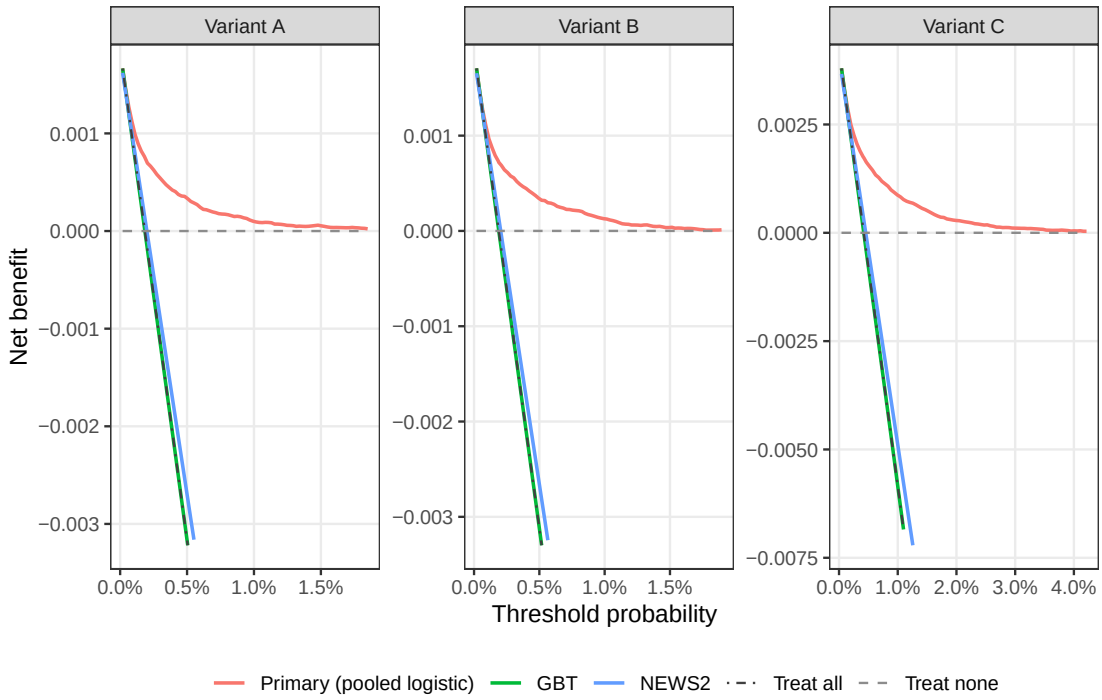


Figure 2: Decision curves on the temporal test set. Net benefit against threshold probability, with both default strategies shown. Curves are clipped below 1.75 times prevalence so the near-zero region stays legible; a line leaving the bottom of a panel is worse than treating nobody from that threshold onward.

0.6 Operating Points at Fixed Sensitivity

For each model the threshold is set to the largest cut-off achieving the target per-hour sensitivity, and PPV, specificity, number needed to evaluate (NNE = 1/PPV), alert rate and time-normalised false-alarm rate are read off at that point.

\begin{table}

\caption{Operating characteristics at a fixed per-hour sensitivity of 80%. NNE = number of alerted person-hours per true event hour. FA/pt-day = false-alert hours per patient-day.}

variant	model	Threshold	Sens	Spec	PPV	NNE	Alerts/pt-day	FA/pt-day	F
A	GBT	0.36600	0.801	0.540	0.00322	311	11.05	11.02	
A	NEWS2	0.15000	0.855	0.301	0.00226	442	16.79	16.75	
A	Primary (pooled logistic)	0.00126	0.801	0.562	0.00338	296	10.52	10.48	
B	GBT	0.38200	0.801	0.511	0.00310	322	11.74	11.70	
B	NEWS2	0.15000	0.858	0.301	0.00232	430	16.79	16.75	
B	Primary (pooled logistic)	0.00128	0.801	0.522	0.00317	315	11.49	11.45	
C	GBT	0.39000	0.800	0.541	0.00733	136	11.04	10.96	
C	NEWS2	0.15000	0.854	0.315	0.00525	191	16.47	16.38	
C	Primary (pooled logistic)	0.00291	0.800	0.565	0.00773	129	10.48	10.40	

\end{table}

Table 2: Net benefit at a threshold probability equal to the event prevalence. Both default strategies have net benefit 0 at this threshold, so Std NB (net benefit divided by the prevalence) is the fraction of the attainable benefit captured. 'Alert rate span' is the change in alert rate across the whole threshold grid: a span of 0 means the score cannot be tuned anywhere in the plausible range and the decision curve collapses onto a default strategy.

variant	model	Threshold	Net benefit	Treat all	Std NB	Alert rate	Alert rate span
A	Cox	0.00185	0.000000	0	0.000	1.000	0.000
A	GBT	0.00185	0.000000	0	0.000	1.000	0.000
A	NEWS2	0.00185	0.000133	0	0.072	0.894	0.000
A	Primary (pooled logistic)	0.00185	0.000753	0	0.407	0.305	0.987
A	qSOFA	0.00185	0.000270	0	0.146	0.656	0.000
A	SIRS	0.00185	0.000251	0	0.136	0.676	0.000
B	Cox	0.00189	0.000000	0	0.000	1.000	0.000
B	GBT	0.00189	0.000000	0	0.000	1.000	0.000
B	NEWS2	0.00189	0.000131	0	0.069	0.894	0.000
B	Primary (pooled logistic)	0.00189	0.000719	0	0.380	0.318	0.995
B	qSOFA	0.00189	0.000284	0	0.150	0.656	0.000
B	SIRS	0.00189	0.000283	0	0.149	0.676	0.000
C	Cox	0.00422	0.000000	0	0.000	1.000	0.000
C	GBT	0.00422	0.000000	0	0.000	1.000	0.000
C	NEWS2	0.00422	0.000298	0	0.071	0.888	0.000
C	Primary (pooled logistic)	0.00422	0.001730	0	0.411	0.289	0.997
C	qSOFA	0.00422	0.000645	0	0.153	0.644	0.000
C	SIRS	0.00422	0.000691	0	0.164	0.665	0.000

Table 3: Per-patient detection at the same thresholds as the operating-point table. Episode sens (6h) = fraction of septic stays alerted inside the actionable window [onset - 6 h, onset]. Episode sens (any pre) = alerted at any point before onset. FA episodes = contiguous runs of alert hours that do not overlap [onset - 6 h, onset + 3 h].

variant	model	Sepsis stays	Episode sens (6h)	Episode sens (any pre)	Patient PPV	%
A	GBT	1015	0.914	0.989	0.0853	
A	NEWS2	1015	0.968	0.987	0.0841	
A	Primary (pooled logistic)	1015	0.880	1.000	0.0837	
B	GBT	1040	0.904	0.987	0.0883	
B	NEWS2	1040	0.967	0.991	0.0865	
B	Primary (pooled logistic)	1040	0.880	0.996	0.0854	
C	GBT	2128	0.902	0.988	0.1810	
C	NEWS2	2128	0.964	0.988	0.1770	
C	Primary (pooled logistic)	2128	0.864	1.000	0.1750	

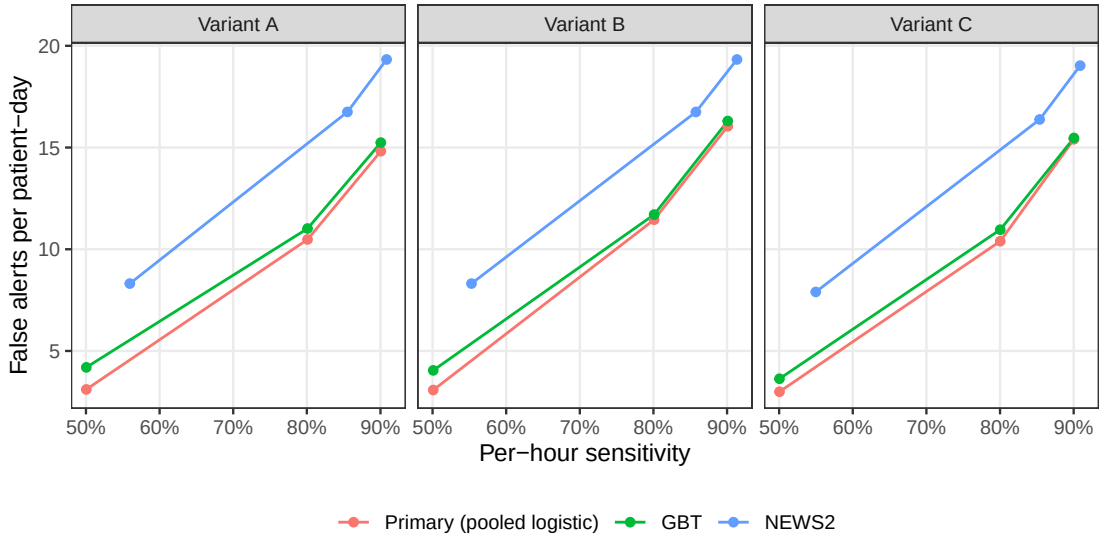


Figure 3: False alerts per patient-day against per-hour sensitivity, at sensitivity targets of 50, 80 and 90 per cent.

0.7 Per-Patient (Event-Level) Detection

Every figure above is per person-hour. Under the discrete-time competing-risks framing, `event == 1` marks the sepsis onset hour itself (states 2 and 3 are discharge alive and death without sepsis), so each septic stay contributes exactly one event hour. Per-hour sensitivity is therefore a strict quantity: the alert must fire *in the onset hour itself*. Per-patient sensitivity is the clinically meaningful counterpart: it asks only whether the patient was alerted at all before onset. The gap between the two, and the price paid for it in false alarms, is the point of this section.

0.7.1 Per-Hour versus Per-Patient, Side by Side

The same operating point read two ways. An hourly false-alarm rate and a per-patient one answer different questions, and a model can look tolerable on the first while being unusable on the second. Both are carried at the headline sensitivity so the gap between them stays visible.

Table 4: Per-hour and per-patient views of the same operating point. Patient-level sensitivity is much higher than per-hour sensitivity because one alert anywhere before onset counts the whole episode as detected; patient-level PPV is correspondingly higher than per-hour PPV because a multi-hour alert run counts once.

variant	model	Sens (hour)	Sens (patient)	PPV (hour)	PPV (patient)	FA hours/pt-c
A	GBT	0.801	0.989	0.00322	0.0853	11
A	NEWS2	0.855	0.987	0.00226	0.0841	16
A	Primary (pooled logistic)	0.801	1.000	0.00338	0.0837	10
B	GBT	0.801	0.987	0.00310	0.0883	11
B	NEWS2	0.858	0.991	0.00232	0.0865	16
B	Primary (pooled logistic)	0.801	0.996	0.00317	0.0854	11
C	GBT	0.800	0.988	0.00733	0.1810	10
C	NEWS2	0.854	0.988	0.00525	0.1770	16
C	Primary (pooled logistic)	0.800	1.000	0.00773	0.1750	10

0.8 Lead Time to Onset

Lead time is the interval between the first alert issued at or before onset and the onset hour itself, computed at the fixed-sensitivity threshold and over the septic stays that were alerted at all.

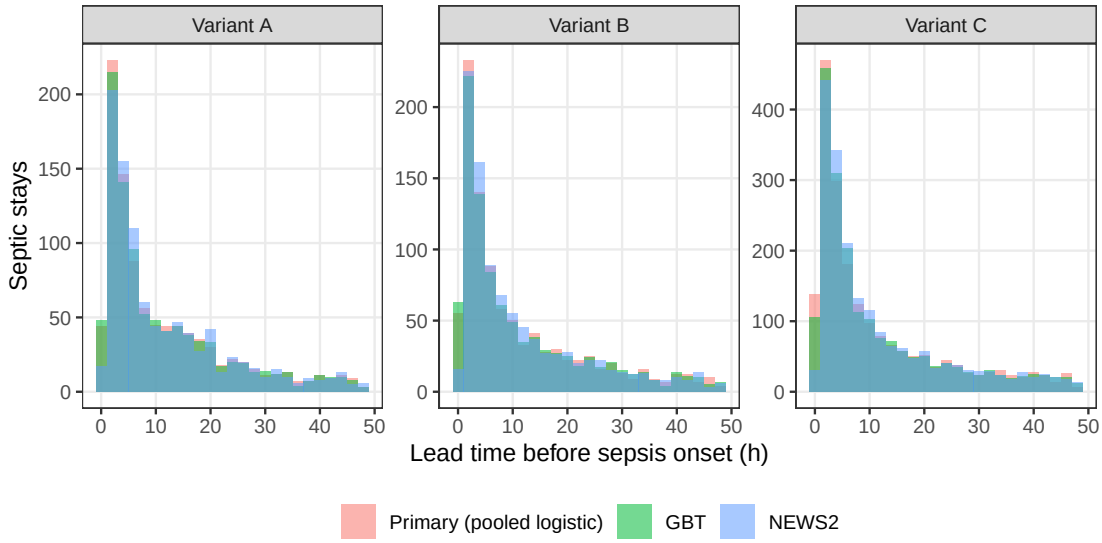


Figure 4: Distribution of lead time before onset at the 80 per cent sensitivity operating point, truncated at 48 hours.

\begin{table}

\caption{Lead time at the 80% sensitivity operating point. '>= 3 h' and '>= 6 h' are fractions of ALL septic stays (not just detected ones) alerted at least that far ahead of onset.}

Table 5: eICU-CRD per-patient detection at the external operating point.

variant	model	n_sepsis_stays	Episode sens (6h)	Episode sens (any pre)	Patient PPV
A	GBT	27	0.852	0.926	0.01010
A	NEWS2	27	0.815	0.852	0.00830
A	Primary (pooled logistic)	27	0.815	1.000	0.00956
B	GBT	29	0.862	0.897	0.00943
B	NEWS2	29	0.862	0.862	0.00889
B	Primary (pooled logistic)	29	0.897	1.000	0.01030
C	GBT	73	0.849	0.877	0.02590
C	NEWS2	73	0.890	0.918	0.02380
C	Primary (pooled logistic)	73	0.877	0.973	0.02520

variant	model	n_sepsis_stays	detected_pre	Median lead (h)	IQR	>= 3 h	>= 6 h
A	GBT	1015	0.989	8	3-19	0.832	0.591
A	NEWS2	1015	0.987	8	4-20	0.868	0.618
A	Primary (pooled logistic)	1015	1.000	8	3-19	0.850	0.593
B	GBT	1040	0.987	8	3-21	0.819	0.579
B	NEWS2	1040	0.991	8	4-21	0.866	0.605
B	Primary (pooled logistic)	1040	0.996	8	3-21	0.827	0.585
C	GBT	2128	0.988	7	3-21	0.822	0.577
C	NEWS2	2128	0.988	8	4-21	0.865	0.605
C	Primary (pooled logistic)	2128	1.000	7	3-21	0.822	0.574

\end{table}

0.9 External Validation: Clinical Utility on eICU-CRD

The same operating-point and per-patient machinery applied to the eICU predictions from notebook 08, so that the clinical-utility picture is available externally as well as internally.

\begin{table}

\caption{eICU-CRD operating characteristics at the same 80% target sensitivity (thresholds re-derived on the external cohort).}

variant	model	Threshold	Sens	PPV	NNE	FA/pt-day
A	GBT	0.38800	0.815	0.0002701	3702	12.06
A	NEWS2	0.10000	0.815	0.0001674	5974	19.46
A	Primary (pooled logistic)	0.00164	0.815	0.0002510	3985	12.98
B	GBT	0.40400	0.828	0.0002039	4905	17.44
B	NEWS2	0.05000	0.828	0.0001665	6004	21.35
B	Primary (pooled logistic)	0.00176	0.828	0.0002965	3372	11.99
C	GBT	0.40700	0.808	0.0005371	1862	16.51
C	NEWS2	0.05000	0.836	0.0004296	2328	21.35
C	Primary (pooled logistic)	0.00462	0.808	0.0007165	1396	12.38

\end{table}

Table 6: Agreement between the dcurves output stored by notebook 07 and the net benefit recomputed here, over the shared threshold range. Differences reflect only the threshold grid offset, not a difference in definition.

variant	model	n_compared	max_abs_diff
A	primary	1	4.00e-07
A	gbt	1	8.31e-05
B	primary	1	1.20e-06
B	gbt	1	7.57e-05
C	primary	4	1.08e-05
C	gbt	4	7.12e-05

0.10 Cross-Check Against Notebook 07

The net-benefit computation here is implemented directly from the Vickers–Elkin formula rather than through `dcurves`. The two are compared at the thresholds they have in common, as a correctness check.

0.11 What These Metrics Add to Discrimination

Everything in this notebook exists because AUROC cannot see it.

Calibration answers whether a predicted risk means what it says. A model can rank patients correctly and still put every probability in the wrong place, which matters the moment anyone picks a threshold to act on.

Net benefit and the operating-point tables answer whether acting on the model beats the two strategies that need no model at all. A curve sitting on top of treat-all is discriminating and contributing nothing.

Lead time answers whether a correct alert arrives early enough to change what happens next. It is also the metric most sensitive to how onset was timed, so it should be read against the label's own construction rather than against published lead times computed under a different onset rule.

Taken together these can disagree with the discrimination ranking, and where they do the disagreement is the finding rather than a problem to reconcile.

0.12 Summary

What this stage leaves behind for stage 12 and for the thesis generator.

```
## Artefacts written to /Users/arif/Desktop/RESEARCH/sepsis/project/output/processed_data :

##   - 11_calibration_summary.parquet
##   - 11_net_benefit.parquet
##   - 11_operating_points.parquet
##   - 11_patient_level_metrics.parquet
##   - 11_hour_vs_patient.parquet
##   - 11_lead_time_summary.parquet
##   - 11_external_operating_points.parquet
##   - 11_external_patient_level.parquet

##
## Figures written to /Users/arif/Desktop/RESEARCH/sepsis/project/output/figures :
```

```
##   - 11_calibration_curves.png
##   - 11_decision_curves.png
##   - 11_false_alarms_vs_sensitivity.png
##   - 11_lead_time_distribution.png
```